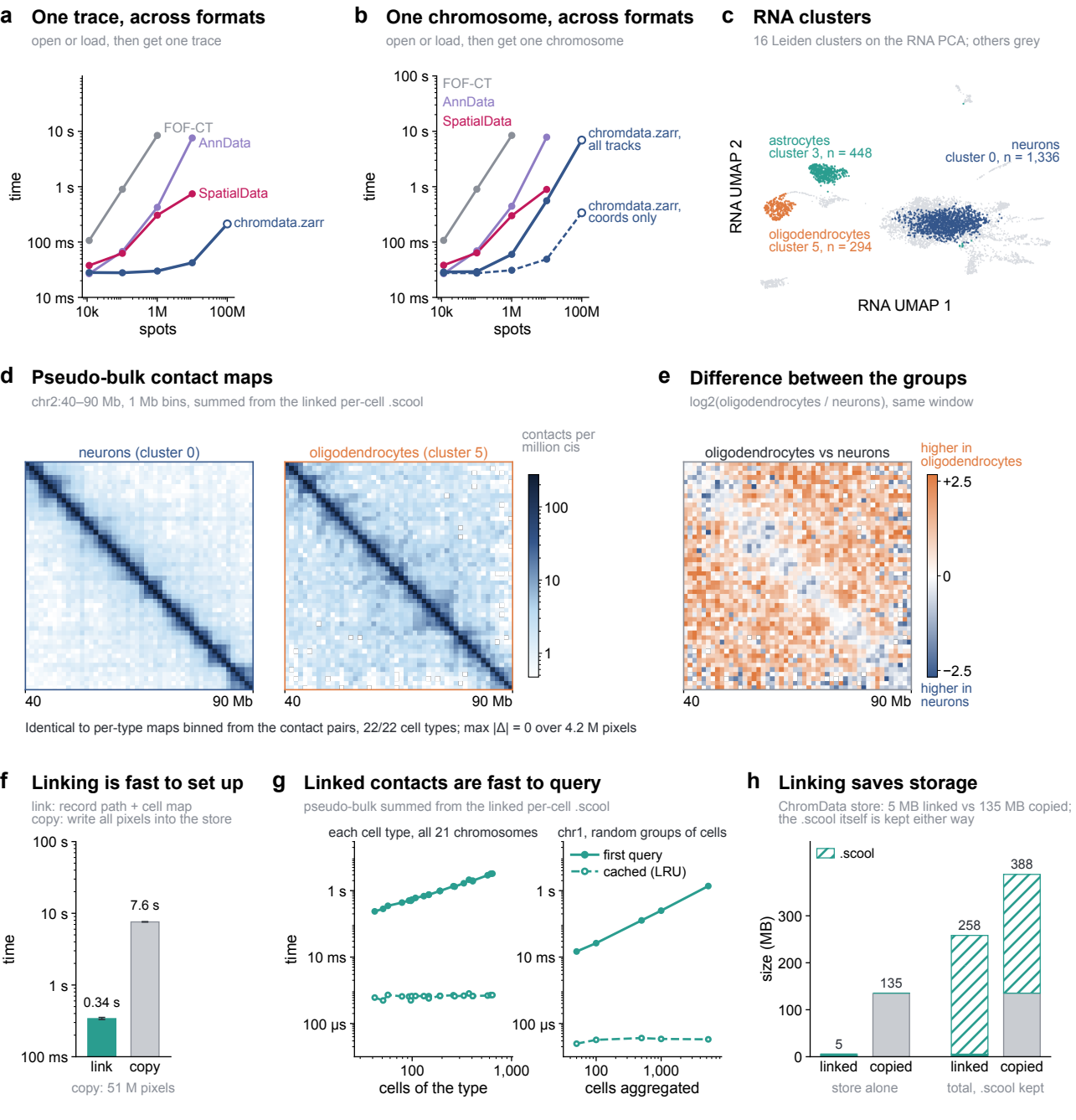

a
One trace, across formats
open or load, then get one trace
time
10 s
1 s
100 ms
10 ms
FOF-CT
AnnData
SpatialData
chromdata.zarr
10k
1M
100M
spots
b
One chromosome, across formats
open or load, then get one chromosome
FOF-CT
AnnData
SpatialData
chromdata.zarr, all tracks
chromdata.zarr, coords only
c
RNA clusters
16 Leiden clusters on the RNA PCA; others grey
RNA UMAP 2
RNA UMAP 1
astrocytes cluster 3, n = 448
neurons cluster 0, n = 1,336
oligodendrocytes cluster 5, n = 294
d
Pseudo-bulk contact maps
chr2:40–90 Mb, 1 Mb bins, summed from the linked per-cell .scool
neurons (cluster 0)
oligodendrocytes (cluster 5)
contacts per million cis
100
10
1
40
90 Mb
e
Difference between the groups
log2(oligodendrocytes / neurons), same window
oligodendrocytes vs neurons
higher in oligodendrocytes
+2.5
0
−2.5
higher in neurons
Identical to per-type maps binned from the contact pairs, 22/22 cell types; max |Δ| = 0 over 4.2 M pixels
f
Linking is fast to set up
link: record path + cell map
copy: write all pixels into the store
7.6 s
0.34 s
link
copy
copy: 51 M pixels
g
Linked contacts are fast to query
pseudo-bulk summed from the linked per-cell .scool
each cell type, all 21 chromosomes
chr1, random groups of cells
first query
cached (LRU)
cells of the type
cells aggregated
h
Linking saves storage
ChromData store: 5 MB linked vs 135 MB copied; the .scool itself is kept either way
.scool
size (MB)
5
135
258
388
linked
copied
store alone
total, .scool kept